

Defence to DDoS Attacks

Module 47

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Video: https://www.youtube.com/watch?v=w_XZu6Rc0jM

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0:00 [Music]

0:15 dear viewers in the previous lecture we have gone through the basics of Doos attacks motivation behind these attacks

0:22 and classification of Dos attacks in today's lecture we will be discussing few aspects of DS attacks that that make

0:30 defending against them difficult after that we will focus on challenges and goals for designing an ideal defense

0:37 solution in the last part of the lecture we will learn some of the tips that can be helpful for preventing the system

0:44 from being attacked by the DS attackers let's begin by learning why dos problem is hard to

0:51 solve dos defense is a challenging issue among the cyber security Community the following characteristics of Dos

0:58 flooding attacks make these attacks very effective for the attacker's purpose and extremely challenging for the defense

1:06 simplified procedures of launching these attacks there are many Doos tools that can be easily downloaded or otherwise

1:13 obtained and set into action they make agent Recruitment and activation automatic and can be used by

1:21 inexperienced users these tools are exceedingly simple and some of them have been around for years still they

1:29 generate effective attacks with little or no tweaking next is traffic variation the similarity of attack traffic to

1:37 legitimate traffics makes separation and filtering extremely hard unlike other security threats that need specially

1:44 crafted packets example intrusions WMS virus flooding attacks need only high traffic volume and can vary packet

1:52 contents and headers value at its will next comes the IP spoofing IP spoofing makes the attack traffic appear as if it

2:01 2 minutes, 1 second comes from numerous legitimate clients this defeats many
resource sharing approaches that identify a client by his
2:09 IP address if IP address were eliminated agents could potentially be distinguished
from legitimate clients by
2:17 their aggressive sending patterns and their traffic could be filtered in the presence
of Ip spoofing the victim sees a lot of service
2:26 initiation request from numerous seemingly legitimate users while the victim could
easily tell those packets apart from ongoing
2:35 Communications with the legitimate users it cannot distinguish new legitimate
requests for service from the attack
2:42 ones thus the victim cannot serve any new users during the attack if the attack is
long the damage
2:50 to the victim's business is obvious another issue is huge volume of traffic the high
volume of attack
2:58 traffic at the victim victim not only overwhelms the targeted resource but makes
traffic profiling hard at such
3:06 high packet rates the defense mechanism can do only simple per packet processing
the main challenge of Dos defense is to
3:13 discern the legitimate from the attack traffic at higher packet speeds next is
several zombies machine
3:22 can be another Factor the strength of adidos attacks lies in numerous agent
machines distributed all over the internet with
3:31 so many agents the attacker can take and even the largest networks and he or she
can vary his or her attack by deploying
3:39 subsets of agents at time or sending few packets from each agent machine wearing
attack strategies defeat many defense
3:47 mechanism that attempt to trace back the attack to its source even the cases when
the attacker does not vary the attacking machines the
3:56 mere number of Agents involved makes Trace back and un attractive solution what
if we knew the identities of 10,000
4:03 machines that are attacking our Network this would hardly get any closer to
stopping the attack the situation would
4:11 clearly be simplified if the attacker were not able to recruit so many agents as
discussed the general increase of internet host and more recently the
4:20 higher percentage of novice internet users suggest that pool of potential agents
will only increase in
4:27 future furthermore the the distributed internet management model makes it
unlikely that any security mechanism will be widely
4:36 deployed thus even if we found ways to secure machines permanently and make
them in previous to the attacker

4:43 intrusion attempts it would take many years until these machines would sufficiently
deploy to impact the Dos

4:51 threat peak spots in the internet topology the current Internet Hub and spoke
topology has a handful of Highly

4:59 connected and very well provisioned spots that relay traffic for rest of the internet
these hubs are highly Provisions to handle heavy traffic in

5:08 the first place but if these few spots were taken down by the attacker or heavily
congested the internet would

5:16 grind to a halt amassing a large number of agent machines and generating heavy
traffic passing through those hotspots

5:23 would have devastating effect on global community so in n sh let's face it a

5:32 flooding based Doos attack seems like a perfect crime in the internet attack tools
and agent machines are abundant

5:39 and easily obtainable a sufficient attack volume is likely to bring the strongest
victim to its knees and the right mixture of

5:47 attack traffic along with the IP spoofing will defeat attack filtering attempts since
numerous business rely

5:56 heavily on the online assess taking that way is show sure to inflict considerable
damage to the victim and finally IP spoofing numerous

6:05 agent machines and lack of automated tracing mechanism across the network
gruny little to no risk to protect of

6:13 being caught next we will discuss about the challenges of DS

6:20 defense the challenges in designing DS defense system fall roughly into two
categories technical challenges and the

6:28 social challenge technical challenges encompasses problems associated with the
current Internet Protocol and characteristics of

6:36 Dos threat Social Challenges on the other hand largely pertain to the manner in
which a successful technical solution

6:45 will be introduced to the internet users and accepted and widely deployed by these
users the main problem that prevails in

6:54 both Technical and social issues is the problem of large scale dos is is a distributed
threat that requires a merit

7:02 of overlapping solutions to various aspects of Dos problem which must be spread
across the internet because

7:09 attacking machines may be spread all over the Internet clearly attack streams can
only be controlled if there is a point of

7:18 Defense between the agents and the victims one approach is to place one defense
system close to the victim so

7:26 that it monitors and controls all the incoming TR traffic this approach has many
deficiencies the main one being

7:34 that system must be able to efficiently handle and process huge traffic volume the other approach is to divide this

7:42 workload by deploying distributed defense defense system must be then deployed in a widespread manner to

7:49 ensure effective action for any combination of agent and victim machines as WID spread deployment cannot

7:58 be granted the technical challenge lies in designing effective defenses that can provide reasonable performance even if

8:05 they are sparsely deployed the social challenge lies in designing an economic model of a defense

8:13 system in a manner that motivates large scale deployment in the internet we will discuss first about the

8:21 technical challenges the distributed nature of DS attacks similarity of the attack packets

8:29 to the legitimate ones and the use of Ip spoofing represents the main technical challenge to the designing effective

8:35 DoS defense system as discussed in the previous section in addition to that advance of DoS defense research has

8:43 historically been hindered by the lack of attack information and absence of standardized evaluation and testing

8:51 approaches the following list summarizes and discusses technical challenges for DS defense

8:59 need for distributed response at many points in the internet there are many possible DoS attacks very few of which

9:08 can be handled only by the victim thus it is necessary to have distributed possibly coordinated response

9:16 system it is also crucial that response be deployed at many points in the internet to cover diverse choices of agents and

9:25 victim since the internet is administrated in a distributed manner wide deployment of any defense system or

9:33 even various system that could cooperate cannot be enforced or granted this discourages many

9:40 researchers from even considering distributed Solutions next comes the lack of detailed attack

9:48 information it is widely believed that reporting occurrences of attack damages the business reputation of victim

9:56 Network therefore very limited information exist about various attacks and incidents are reported only to government organizations under

10:04 obligation to keep them secret it is difficult to design an imaginative solution to the problem if

10:11 one cannot become familiar with it note that the attack information should not be
confused with the attack tool information which is publicly available

10:20 at many internet sites attack information would include the attack type time and
duration of attack number

10:28 of Agents involved if this information is known attempted response and its
Effectiveness and damage is

10:36 suffered lack of defense system benchmarks many vendors make bold claims that
their solution completely handles

10:44 the deos problem there is currently no standardized approach for testing dos
defense system that would enable their

10:52 comparison and characterization this has two detal influences on dos research

11:00 since there is no attack Benchmark defense designers are allowed to present those
tests that are most advantageous

11:08 to their system and researchers cannot compare actual performance of their
solution to the existing defenses

11:17 instead they can only comment on design issues next comes the difficulty of large
scale

11:25 testing Doos defense need to be tested in a realistic enir environment this is
currently impossible due to the lack of

11:33 large scale test beds safe ways to perform live distributed experiments across the
internet or detailed and

11:40 realistic simulation tools that can support several thousands of nodes Claim about
defense system

11:48 performance are thus made based on small scale experiments or simulations are
not

11:55 credible some test bits are in use right now by DS resers example Planet lab and
IMU

12:03 lab now we will discuss about the Social Challenges many dos defense systems
require certain deployment patterns to

12:12 be effective those patterns fall into several category complete deployment a
given system is deployed at each host router or network in the internet next is

12:20 continuous deployment a given system

12:29 is deployed at host or routers that are directly connected next is large scale
widespread deployment the majority of

12:38 hosts in the internet deploy a given System complete deployment at specified
points in the

12:46 internet there is a set of carefully selected deployment points All Points must
deploy the proposed defense to

12:53 achieve the desired security modification of widely deployed internet protocols
such as tcpip or

13:03 HTTP next we will discuss that what are the goals necessary to have a strong and effective defense

13:11 system following are some of Doos defense goals which Doos defense strategies strive to

13:19 achieve first is Effectiveness a good Dos defense should actually defend it should provide either

13:28 effective prent prevention that really makes attacks impossible or effective reaction ensuring that dos effect goes

13:37 away in the case of reactive mechanism the response should be sufficiently quick and automated to ensure that the

13:44 target does not suffer seriously from the attack next goal should be completeness a good Dos defense should

13:53 handle all possible attacks if that degree of perfection is impossible it should at least handle a large number of

14:02 them a mechanism that is capable of handling an attack based on TCP sin flooding but cannot offer any assistance

14:10 if a ping flood arrives is clearly less valuable than a defense that can handle both styles of

14:18 attack thus a preventive measures like TCP sin cookies helps but is not

14:25 sufficient unless coupled with other defense mechanisms completeness is also required in detection and reaction if our detection

14:35 mechanism does not recognize a particular pattern of incoming packet as an attack pre assumably it will not work

14:42 invoke any response and attack will succeed while completeness is an obvious goal it is extremely hard to

14:51 achieve since attackers are likely to develop new types of attacks specially designed to bypass existing defense

14:59 moreover flash events discussed need clear-cut distinctions from dos attacks next is minimum collateral

15:09 damage as discussed above the core goal of Doos defense is not to stop dos attack packets but to ensure that

15:17 legitimate users can continue to perform their normal activities despite the presence of Dos attack clearly a good

15:25 defense mechanism must achieve it some legitimate traffic may be flowing from sides that are also sending attack

15:33 traffic other legitimate traffic is destined for nodes on same network as the target node there may be a

15:41 legitimate traffic that is neither coming from an attack machine nor being delivered to the targets network but perhaps shares some portion of its path

15:50 through the internet with some of the attack traffic and some of the legitimate traffic measures other characteristics

15:58 with the attack traffic such as application protocol or destination Port potentially making it difficult to

16:05 distinguish between them none of these legitimate traffic categories should be distributed by the

16:12 DoS defense mechanism however DoS defense mechanisms are not able to characterize DoS traffic accurately and

16:21 drop a part of legitimate traffic also this dropping of legitimate traffic is called collateral damage

16:30 since DoS attackers often strive to conceal their attack traffic in the legitimate traffic stream it is common

16:38 for legitimate traffic to closely resemble the attack packets so the problem of collateral damage is real and

16:46 serious especially in case of flash events protecting legitimate traffic that is minimizing collateral

16:54 damage is very important next comes the low False positive rates defense mechanism should Target only

17:02 true DoS attacks preventive mechanism should not have the effect of hurting other forms of net

17:09 traffic reactive mechanism should be activated only when a DoS attack is actually

17:16 underway a detection scheme can falsely detect an attack in a situation when no attack was actually

17:23 happening this is called false positive these false positive CA collateral damage as they trigger the

17:31 filtering mechanism when there is actually no attack moreover reactive mechanisms are likely to incur some

17:39 cost the cost is in terms of extra CPU Cycles memory and delay caused on processing of all

17:47 packets low deployment and operational cost DoS defense are meant to allow systems to continue operations during DoS

17:56 attacks the cost associated with the defense system must be commensurate with Benefits provided by it for

18:04 commercial Solutions there is an obvious economic cost of buying the hardware and software required to run it usually

18:12 there are also significant system administration cost with setting up new security equipment or

18:20 software depending upon the character of DoS defense mechanism it may require frequent ongoing

18:27 Administration for example a mechanism based on detecting signatures of particular attacks will need to

18:34 receive update as new attacks are characterized requiring either manual or automated actions other operational cost

18:43 relate to overheads imposed by the defense system A system that performs stateful

18:50 inspection of all incoming packets May delay each packet for example or a system that throttles data streams from

18:59 suspicious sources may slow down any legitimate interactions with those sources unless such cost are extremely

19:07 low or extremely rare paid they must be balanced against benefits of achieving some degrees of protection against dos

19:17 attacks how we can defend against these attacks our next part of discussion will focus on the

19:25 defense Doos defense modules are shown in the Fig figure generally there are four broad

19:32 categories of defense against dos attacks first is attack prevention second is attack detection

19:40 third is attack Source identification and fourth is attack reaction attack preventions aims to stop

19:49 attacks before they can reach their target it refers to the filtering spoof packets close to or at the attack

19:57 sources which is one of the most effective defense approaches for dos attacks that use poofed

20:05 traffic attack detection aims to detect dos attacks when they occur attack detection is an important

20:12 procedures to direct any further action it is the process of detecting the attack based on the traffic feature

20:21 distribution signature based detection techniques are not applied for dos defense due to everchanging nature of signature of

20:30 attacks anomaly based detection techniques are suitable for for this purpose next is attack Source

20:38 identification which aims to locate the attack sources regardless of whether the source address field in which packet

20:45 contains enormous information it is a crucial steps to minimize the attack damage and provide

20:53 prevention to the potential attackers next comes the attack reactions with which aims to eliminate

21:00 or curtail the effects of an attack it is the final step in defending against dos attacks and therefore determines the

21:08 overall performance of the defense mechanism the challenge of the attack reaction is however to filter and attack traffic without disturbing the

21:16 legitimate traffic in the last part of the discussion we will learn some of the important tips which helps us to protect

21:25 our system from being attacked by dos first is Implement router filters this will lessen your exposures to certain

21:34 denial of service attacks additionally it will Aid in preventing users on your network from effectively launching

21:42 certain denial of service attacks second if they are available for your system install patches to guard against TCP sin

21:51 flooding this will substantially reduce your exposure to those attacks but may not eliminate the risk entirely

22:00 third is disable any unused or unneeded Network Services this can limit the ability of an intruder to take advantage

22:08 of those services to execute a denial of service attack next is enable quota systems on

22:15 your operating system if they are available for example if your operating system supports dis quotas enable them

22:23 for all accounts especially accounts that operate Network Services in addition if your operating system

22:30 support partitions or volume consider partitioning your file system so as to separate critical functions from other

22:39 activities next is observe your system performance and establish baselines for ordinary

22:47 activity next use the baselines to cause unusual levels of disk activity CPU uses

22:54 or network traffic next routinely examine your physical security with respect to your

23:02 current needs consider servers routers un attending terminals network access points wiring closets environmental

23:11 systems such as air and power and other components of your system next use trip wire or similar

23:19 tool to detect changes in the configuration information or other files next invest in redundant and fall

23:28 TN Network configurations establish and maintain regular backup schedules and policies

23:35 particularly for important configuration information next establish and maintain

23:42 appropriate password policies especially assessed to highly privileged account such as Unix rout or Microsoft Windows

23:52 administration at last I would like to conclude that DS attacks is a devastating attack among the cyber community and the defense to this attack

24:01 24 minutes, 1 second is eluded so far though there are many academic and Commercial Solutions present in the

24:08 market in this lecture we have discussed some of the factors that made dos attack problem harder to solve we have also

24:15 learned some of the key challenges and goals for designing an ideal defense solution some tips to manage your system against these attacks are also

24:24 highlighted thank you all

24:27 [Music]

24:48 [Music]